Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

AI

Section Id :	640653138842
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	11
Number of Questions to be attempted :	11
Section Marks :	32
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653387301
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532126169 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR

PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536928538. ✔ Yes

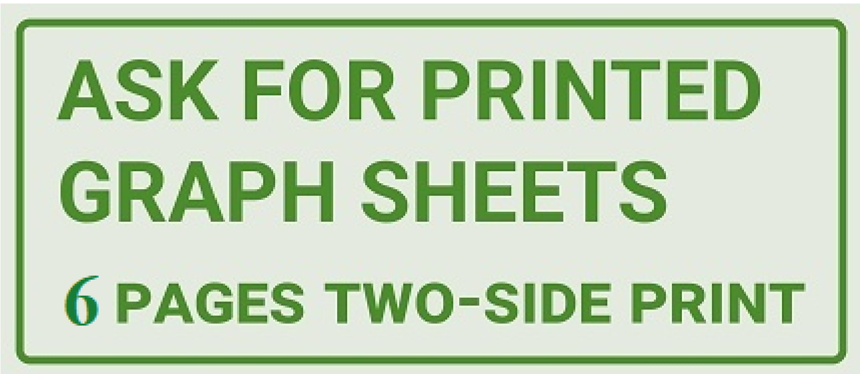
6406536928539. ✖ No

Sub-Section Number :	2
Sub-Section Id :	640653387302
Question Shuffling Allowed :	No

Question Number : 2 Question Id : 6406532126170 Question Type : MSQ

Correct Marks : 0 Max. Selectable Options : 0

Question Label : Multiple Select Question



Options :

6406536928540. ✔ Printed graph sheets were provided on time.

6406536928541. ✖ Printed graph sheets were provided late.

6406536928542. ✖ Printed graph sheets were not provided.

6406536928543. ✔ I used the graph sheets.

6406536928544. ✖ I did not use graph sheets.

Sub-Section Number :	3
Sub-Section Id :	640653387303
Question Shuffling Allowed :	No

Question Id : 6406532126171 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (3 to 4)

Question Label : Comprehension

SEARCH

Consider a water-jug puzzle with three jugs a, b and c of capacities 5L, 3L and 2L, respectively.

A state, encoded as a digit string ABC, denotes volume of water in jugs a, b and c, respectively.

For example, state 122 denotes that jugs a, b and c contain 1L, 2L and 2L of water, respectively.

The allowable moves (actions) are

xETy: pick up jug x and empty it into jug y and jug y is also topped up,
xEy: pick up jug x and empty it into jug y and jug y is still not topped up,
xTy: pick up jug x and top up jug y and jug x is still not emptied out.

For example: 122 ---cEa---> 320 ---bETa---> 500 ---aTb---> 230.

MoveGen takes a state ABC and returns a set of neighbours that are one move away from ABC.

For example, MoveGen(122) = { 032, 302, 320, 131 }.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 3 Question Id : 6406532126172 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Let 212 be the start state and 131 be the goal state, find the shortest path from start to goal. Enter the path starting with 212 as a comma separated list of states.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.

Answer format: 212,230,500

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

212,410,401,131

Question Number : 4 Question Id : 6406532126173 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Let 212 be the start state and 131 be the goal state, find the sequence of moves that produce the shortest path from start to goal. Enter the sequence of moves as a comma separated list.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.
Answer format: aEb,aTc,bETa,cETb
Response Type : Alphanumeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Answers Case Sensitive : No
Text Areas : PlainText
Possible Answers :

cEa,bEc,aTb

Sub-Section Number :	4
Sub-Section Id :	640653387304
Question Shuffling Allowed :	No

Question Id : 6406532126174 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (5 to 7)
Question Label : Comprehension
SEARCH ALGORITHMS

Answer the given subquestions.
Sub questions

Question Number : 5 Question Id : 6406532126175 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
Under what cases does Depth First Iterative Deepening (DFID) guarantee to find the shortest path if one exists?
Options :
6406536928547. ✖ When it inspects only new nodes.
6406536928548. ✖ When it inspects new as well as open nodes.
6406536928549. ✔ When it inspects new as well as closed nodes.
6406536928550. ✖ None of these

Question Number : 6 Question Id : 6406532126176 Question Type : MSQ
Correct Marks : 1 Max. Selectable Options : 0
Question Label : Multiple Select Question
Given a finite state space with unit edge costs, and a heuristic function whose properties are not known, which of the following algorithms are suitable for finding the optimal path?
Options :
6406536928551. ✖ Depth First Search
6406536928552. ✔ Breadth First Search

6406536928553. ✔ Dijkstra's algorithm

6406536928554. ✔ Branch and Bound Search

6406536928555. ✖ A*

6406536928556. ✖ WA* for some w

6406536928557. ✖ Sparse Memory Graph Search (SMGS)

Question Number : 7 Question Id : 6406532126177 Question Type : MSQ
Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

If w is set to a large value (tending to infinity) then wA* algorithm will _________ .

Options :

6406536928558. ✔ behave like Best-First Search

6406536928559. ✖ behave like Dijkstra's algorithm

6406536928560. ✖ guarantee an optimal path if h is admissible

6406536928561. ✔ not guarantee an optimal path even if h is admissible

Sub-Section Number :	5
Sub-Section Id :	640653387305
Question Shuffling Allowed :	No

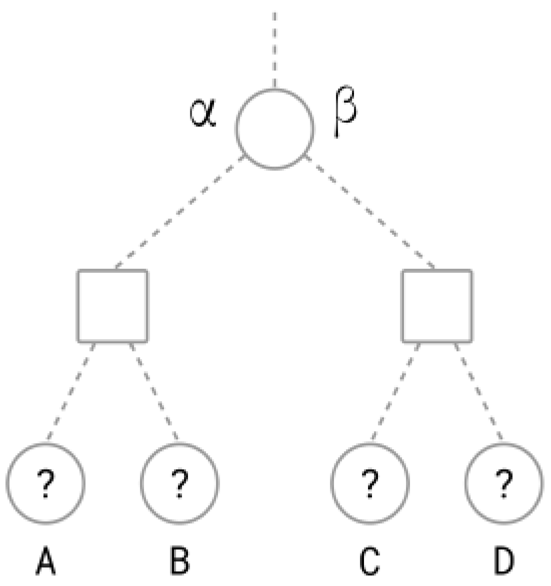
Question Id : 6406532126178 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (8 to 11)

Question Label : Comprehension

GAMES: ALPHA-BETA

Consider a game tree with the root node as MAX, where an arbitrary path from the root reaches the subtree shown in the figure.



Each leaf node A to D takes a **UNIQUE EVAL VALUE** from the set {1, 2, 3, 4}.

Alpha-Beta algorithm is entering the subtree with alpha=1 and beta=3;
find an optimal eval assignment (for nodes A to D) that maximizes the number of leaves pruned;
find the minimax value, the type of cuts and the leaves pruned for that assignment.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 8 Question Id : 6406532126179 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Enter the optimal eval assignment (evals of nodes A to D) in the text box.

Enter a comma separated list of evals.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.
Answer format: 1,2,3,4

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

3,1,4,2

3,2,4,1

4,1,3,2

4,2,3,1

Question Number : 9 Question Id : 6406532126180 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

The minimax value of the subtree for the optimal eval assignment is __________ .

Enter an integer.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 10 Question Id : 6406532126181 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Enter the number of alpha-cuts followed by the number of beta-cuts as a comma separated list.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.

Answer format: 1,2

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

0,2

Question Number : 11 Question Id : 6406532126182 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Enter the label of leaf nodes pruned by Alpha-Beta algorithm, or enter NIL if no leaves were pruned.

Enter a comma separated list of labels (A to D), or enter NIL.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.

Answer format: W,X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

B,D

D,B

Sub-Section Number :	6
Sub-Section Id :	640653387306
Question Shuffling Allowed :	No

Question Id : 6406532126183 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

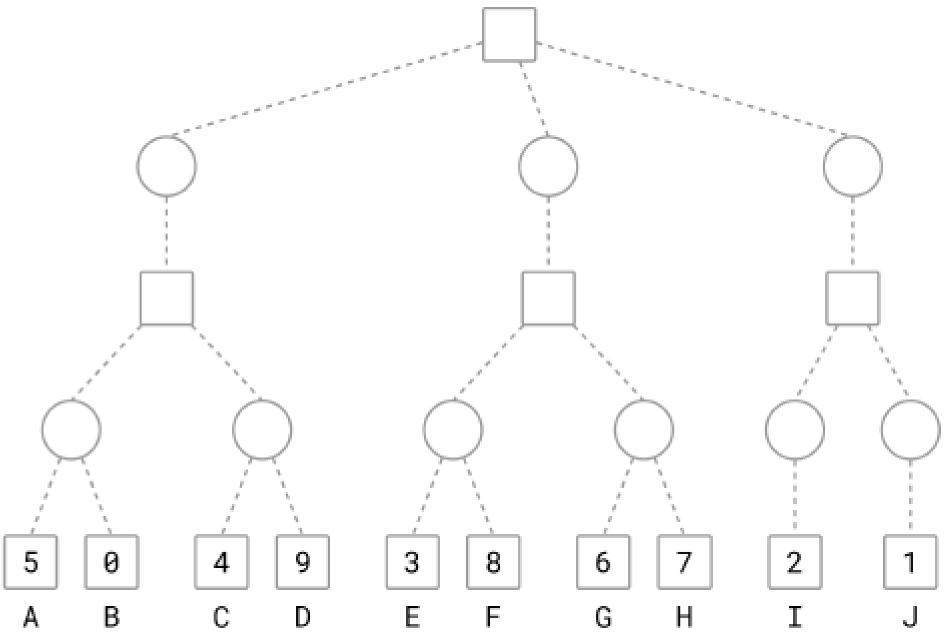
Question Numbers : (12 to 13)

Question Label : Comprehension

GAMES: SSS STAR

The figure shows a game tree with evaluation function values at the horizon nodes.
The horizon nodes are labeled from A to J.
Where applicable, use these labels in short answers.

Tie-breaker:
When several nodes carry the same best cost then select the deepest node, if tie persists then select the leftmost of the deepest nodes to break the tie.



Run SSS* algorithm on the game tree, then answer the sub-questions.

Sub questions

Question Number : 12 Question Id : 6406532126184 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Find the horizon nodes that are assigned SOLVED status by SSS* algorithm. Enter the labels of those nodes in the textbox, or enter NIL if no nodes are assigned SOLVED status.

Enter node labels as a comma separated list in **alphabetical** order.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,C,E,G,H,I,J

A,C,E,G,I,J,H

Question Number : 13 Question Id : 6406532126185 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Find the SOLVED horizon nodes that are pruned from the queue by SSS* algorithm, i.e., SOLVED horizon nodes removed from the queue when a MAX-ancestor is SOLVED. Enter the labels of those nodes in the textbox, or enter NIL if SOLVED nodes were never pruned.

Enter node labels as a comma separated list in **alphabetical** order.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,E,I,J

Sub-Section Number :	7
Sub-Section Id :	640653387307
Question Shuffling Allowed :	No

Question Id : 6406532126186 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (14 to 16)

Question Label : Comprehension

PROBLEM DECOMPOSITION

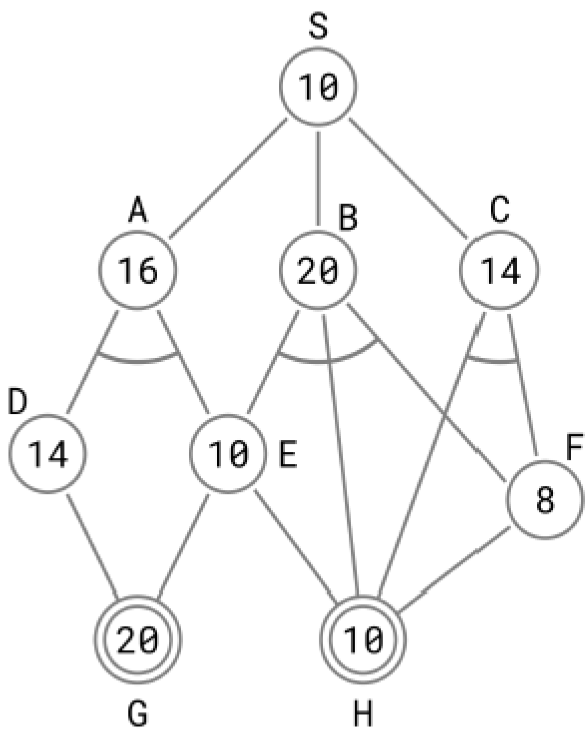
The figure shows an AND-OR decomposition of problem S into smaller problems.
The nodes are uniquely identified by labels (S, A, B, C, ...).

Each node shows the heuristic estimate of the cost of solving that node.
Nodes shown in double lines are primitive nodes and their values are actual costs.

A primitive node is added to the graph, with SOLVED status, when its parent is expanded. And therefore, a primitive node is never expanded.

The cost of each edge is 2 units.

Tie-breaker 1: If several nodes have the same cost then break the tie using node labels.
Tie-breaker 2: For AND nodes, select the unsolved branch with the highest cost.



Use AO* algorithm to solve S, then answer the sub-questions.
Sub questions

Question Number : 14 Question Id : 6406532126187 Question Type : SA Keyboard Layout : Inscript
Correct Marks : 1
Question Label : Short Answer Question
List the nodes expanded by AO* algorithm. List the nodes in the order they are expanded.
Observe that primitive nodes are not expanded.

Enter a comma separated list of node labels.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.
Answer format: S,X,Y,Z
Response Type : Alphanumeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Set
Answers Case Sensitive : No
Text Areas : PlainText
Possible Answers :

S,C,A,B,F

C,A,B,F

Question Number : 15 Question Id : 6406532126188 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

For each node expanded by AO* algorithm, determine the value propagated to the start node S. Enter the values of S as a list.

Enter a comma separated list of numbers.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.
Answer format: 12,42,17

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

16,18,22,24,28

18,22,24,28

Question Number : 16 Question Id : 6406532126189 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What can you conclude about the given AND-OR decomposition?

Options :

6406536928570. ✔ The heuristic is admissible.

6406536928571. ✖ The heuristic is inadmissible.

6406536928572. ✖ The heuristic is sometimes admissible and sometimes inadmissible.

Question Id : 6406532126190 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

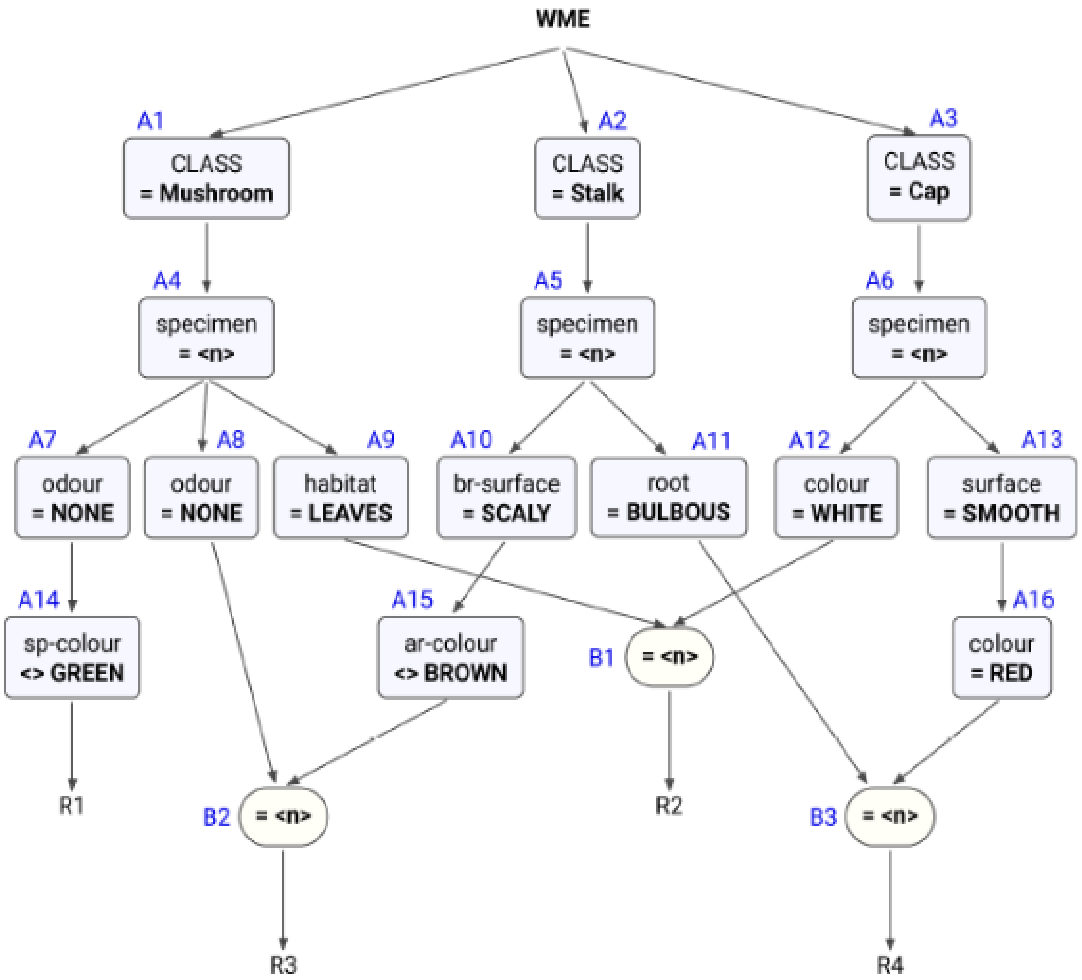
Question Numbers : (17 to 19)

Question Label : Comprehension

RULE BASED EXPERT SYSTEMS

A part of the Rete Net that classifies mushrooms (as edible or poisonous) is shown in the figure. The labels A1, A2, ..., A10, A16, ..., B1, B2, B3, R1, ..., R4 uniquely identify the nodes in the network. When required, use the above label ordering to **break ties** and to enter short answers.

Note: "attribute <> x" is true when that attribute exists and its value is not equal to x.



Run the Rete algorithm for the Working Memory shown below, the WMEs are in timestamp order. Assume that WMEs reside at appropriate Alpha nodes, and the Beta nodes point to WMEs residing in Alpha nodes.

```
101.  (Cap ^specimen C36  ^colour RED  ^surface SMOOTH)
102.  (Cap ^specimen A25  ^colour WHITE  ^surface SMOOTH)
103.  (Mushroom ^specimen X16  ^odour NONE  ^habitat LEAVES)
104.  (Mushroom ^specimen A25  ^odour NONE  ^habitat LEAVES)
105.  (Stalk ^specimen C36  ^root BULBOUS  ^ar-colour WHITE)
106.  (Stalk ^specimen X16  ^br-surface SCALY  ^ar-colour WHITE)
107.  (Mushroom ^specimen C36  ^odour NONE  ^sp-colour WHITE)
108.  (Mushroom ^specimen B49  ^odour ALMOND  ^sp-colour BROWN)
109.  (Stalk ^specimen B49  ^br-surface SMOOTH)
```

For each WME identify its location (node label) in the Rete Net, and prepare the conflict set for the first cycle, then answer the sub-questions.

Sub questions

Question Number : 17 Question Id : 6406532126191 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following rule-data tuples are in the conflict-set?

Options :

6406536928573. ✔ R1,107

6406536928574. ✔ R2,102,104

6406536928575. ✔ R3,103,106

6406536928576. ✔ R4,101,105

6406536928577. ✖ R2,102,103

6406536928578. ✖ R3,104,106

Question Number : 18 Question Id : 6406532126192 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

If the Inference Engine uses **Specificity** as the conflict resolution strategy then which of the following rule-data tuples will qualify?

Options :

6406536928579. ✖ R1,107

6406536928580. ✖ R2,102,104

6406536928581. ✔ R3,103,106

6406536928582. ✔ R4,101,105

6406536928583. ✖ R2,102,103

6406536928584. ✖ R3,104,106

Question Number : 19 Question Id : 6406532126193 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Recency** as the conflict resolution strategy then which of the following rule-data tuples will qualify?.

Options :

6406536928585. ✔ R1,107

6406536928586. ✖ R2,102,104

6406536928587. ✖ R3,103,106

6406536928588. ✖ R4,101,105

6406536928589. ✖ R2,102,103

6406536928590. ✖ R3,104,106

Question Id : 6406532126194 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (20 to 22)

Question Label : Comprehension

AUTOMATED PLANNING 1

Answer the given subquestions.
Sub questions

Question Number : 20 Question Id : 6406532126195 Question Type : MSQ
Correct Marks : 1 Max. Selectable Options : 0
Question Label : Multiple Select Question
Consider actions a and b and the two orderings (a then b) and (b then a). Which of the following conditions (each taken independently) will make both of the orderings feasible?
Options :
6406536928591. ✔ P is in pre(a) and also in pre(b).
6406536928592. ✔ P is in del-effects(a) and also in del-effects(b).
6406536928593. ✖ None of these.

Question Number : 21 Question Id : 6406532126196 Question Type : MSQ
Correct Marks : 1 Max. Selectable Options : 0
Question Label : Multiple Select Question
In planning graphs constructed by GraphPlan, actions a and b in layer n are mutex __________ .
Options :
6406536928594. ✔ if P in pre(a) and Q in pre(b) are mutex
6406536928595. ✔ if P is in pre(a), in del-effects(a), in pre(b) and also in del-effects(b)
6406536928596. ✖ only if every P in pre(a) is also present in del-effects(b)
6406536928597. ✖ only if every P in add-effects(a) is also present in del-effects(b)

Question Number : 22 Question Id : 6406532126197 Question Type : MSQ
Correct Marks : 1 Max. Selectable Options : 0
Question Label : Multiple Select Question
In planning graphs constructed by GraphPlan, which of the following are true?
Options :
6406536928598. ✔ If propositions P and Q are non mutex in layer zero then it will remain non mutex in all future layers.
6406536928599. ✔ If propositions P and Q are non mutex in a layer n then it will remain non mutex in all future layers.
6406536928600. ✖ If propositions P and Q are non mutex in a layer n then it can become mutex in a future layer.
6406536928601. ✖ None of these.

Sub-Section Number :	8
Sub-Section Id :	640653387308
Question Shuffling Allowed :	No

Question Id : 6406532126198 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (23 to 29)
Question Label : Comprehension
AUTOMATED PLANNING 2

The domain description of a Blocks World with a single one-armed robot is given below.

```
PREDICATES

armEmpty        The arm is not holding any block, it is empty.
holding(X)      The arm is holding X.
onTable(X)      X is on the table.
clear(X)        X has nothing above it, it is clear.
on(X,Y)         X is directly placed on Y.

OPERATORS

Pickup(X): pick up X from the table.
     Preconditions: { armEmpty, clear(X), onTable(X) }
     Add Effects  : { holding(X) }
     Del Effects  : { armEmpty, onTable(X) }

Putdown(X): place X on the table.
     Preconditions: { holding(X) }
     Add Effects  : { armEmpty, onTable(X) }
     Del Effects  : { holding(X) }

Unstack(X,Y): pick up X that is directly sitting on Y.
     Preconditions: { armEmpty, clear(X), on(X,Y) }
     Add Effects  : { clear(Y), holding(X) }
     Del Effects  : { armempty, on(X,Y) }

Stack(X,Y): place X directly on top of Y.
     Preconditions: { holding(X), clear(Y) }
     Add Effects  : { armEmpty, on(X,Y) }
     Del Effects  : { holding(X), clear(Y) }

NOTE: variables X and Y must bind to different objects.
```

Consider the following planning problem.

```
Start State: { onTable(A), onTable(B),
               on(C,A), on(E,B),
               clear(C), clear(E), clear(D),
               holding(D)
             }

Goal Description: { on(D,E), on(A,D) }
```

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 23 Question Id : 6406532126199 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions for FSSP?

Options :

6406536928602. ✔ Putdown(D)

6406536928603. ✖ Stack(A,D)

6406536928604. ✔ Stack(D,E)

6406536928605. ✔ Stack(D,C)

6406536928606. ✖ Unstack(E,B)

6406536928607. ✖ Unstack(C,A)

Question Number : 24 Question Id : 6406532126200 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **relevant** actions for BSSP?

Options :

6406536928608. ✖ Putdown(D)

6406536928609. ✔ Stack(A,D)

6406536928610. ✔ Stack(D,E)

6406536928611. ✖ Stack(D,C)

6406536928612. ✖ Unstack(E,B)

6406536928613. ✖ Unstack(C,A)

Question Number : 25 Question Id : 6406532126201 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In the planning graph, which of the following are mutex action pairs in layer 1?

Options :

6406536928614. ✖ Stack(D,E) and nop for armEmpty

6406536928615. ✖ Stack(D,E) and nop for clear(C)

6406536928616. ✔ Stack(D,E) and nop for holding(D)

6406536928617. ✔ Stack(D,E) and Putdown(D)

Question Number : 26 Question Id : 6406532126202 Question Type : MSQ
Correct Marks : 1 Max. Selectable Options : 0
Question Label : Multiple Select Question
In the planning graph, which of the following are mutex proposition pairs in layer 1?
Options :

6406536928618. ✖ armEmpty and clear(C)

6406536928619. ✔ armEmpty and holding(D)

6406536928620. ✖ on(D,E) and clear(C)

6406536928621. ✔ on(D,E) and onTable(D)

Question Number : 27 Question Id : 6406532126203 Question Type : MSQ
Correct Marks : 1 Max. Selectable Options : 0
Question Label : Multiple Select Question
In the planning graph, which of the following are **applicable** actions in layer 2?
Options :

6406536928622. ✔ Pickup(D)

6406536928623. ✖ Putdown(C)

6406536928624. ✔ Unstack(E,B)

6406536928625. ✔ Unstack(C,A)

Question Number : 28 Question Id : 6406532126204 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
For the given planning problem, does a plan exist?
Options :

6406536928626. ✔ Yes

6406536928627. ✖ No

6406536928628. ✖ Cannot be determined

Question Number : 29 Question Id : 6406532126205 Question Type : SA Keyboard Layout : Inscript
Correct Marks : 1
Question Label : Short Answer Question
For the given planning problem, the length of the plan found by GraphPlan is __________ .

Enter an integer or enter NIL if GraphPlan cannot find a plan.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.
Answer format: 12

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number :	9
Sub-Section Id :	640653387309
Question Shuffling Allowed :	No

Question Id : 6406532126206 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (30 to 34)

Question Label : Comprehension

CONSTRAINT SATISFACTION

Consider a CSP over 3 variables A, B, C where the domains and constraints are:

$D_A = D_B = D_C = \{1,2,3,4\}$

R_AC: A = mod(2C,3)

R_BA: B = mod(2A,7)

R_CB: C = mod(2B,7)

where, mod(n,d) returns the remainder after dividing n by d, for example, mod(8,3)=2, mod(6,7)=6, mod(8,7)=1.

Compute the three relations to match the domain constraints,then draw the constraint graph and matching-diagram,and then answer the sub-questions.

Sub questions

Question Number : 30 Question Id : 6406532126207 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following tuples occur in R_AC?

Options :

6406536928630. ✔ (1,2)

6406536928631. ✔ (2,1)

6406536928632. ✔ (2,4)

6406536928633. ✖ (4,2)

Question Number : 31 Question Id : 6406532126208 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following tuples occur in R_BA?

Options :

6406536928634. ✔ (1,4)

6406536928635. ✔ (2,1)

6406536928636. ✖ (3,1)

6406536928637. ✔ (4,2)

Question Number : 32 Question Id : 6406532126209 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Is the given CSP arc-consistent?

Options :

6406536928638. ✖ Yes

6406536928639. ✔ No

6406536928640. ✖ Cannot be determined

Question Number : 33 Question Id : 6406532126210 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the given CSP is not already arc-consistent, then make it arc-consistent, and then check if the resulting network is path consistent.

Options :

6406536928641. ✔ It is path consistent.

6406536928642. ✖ It is not path consistent.

6406536928643. ✖ Path consistency does not apply because the constraint graph has a cycle.

Question Number : 34 Question Id : 6406532126211 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Does the given CSP have a solution?

Enter the solution for the variables A,B,C as a comma separated list.
Enter NIL if there is no solution.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.

Answer format: 1,2,3

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

2,4,1